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FEDERAL SPECIFICATION

PIPE, CORRUGATED (ALUMINUM ALLOY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers corrugated aluminum alloy pipe for use as culverts, storm drains, underdrains, conduits, and miscellaneous applications.

1.2 Classification.

1.2.1 Class, series, and shape. Pipe, pipe arches, and arches shall be of the following classes, series (see table II), and shape as specified (see 6.2):

Class I - Pipe, annular corrugations, Series A and B
II - Pipe, helical corrugations, Series A and C
III - Structural plate pipe, annular corrugations, Series D, bolted field assembly

Shape 1 - Pipe, full circular cross section
2 - Pipe, factory elongated
3 - Pipe arch cross section
4 - Arches of circular arc cross section

1.2.2 Coating. The following coatings are available to provide additional protection in unusually severe environments. The appropriate coating, when required, shall be specified (see 6.2).

Coating A - Fully bituminous coated
B - Half bituminous coated, paved invert
C - Fully bituminous coated, paved invert
D - Paved invert only
E - Fully bituminous coated, fully paved

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-781 - Steel Strapping, Flat
PPP-B-601 - Boxes, Wood, Cleated, Plywood
PPP-B-636 - Box, Fiberboard
PPP-C-650 - Crates, Wood, Open and Covered

FSC 4710

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agency).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Test Method Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specification:

MIL-P-236 - Pipe, Culvert, Nestable.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this Specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Publications:

A153 - Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
A449 - Specification for Quenched and Tempered Steel Bolts and Studs.
A536 - Specification for Carbon Steel Nuts.
A307 - Specification for Low Carbon Steel Externally and Internally Threaded Standard Fasteners.
A325 - Specification for High Strength Steel Bolts.
A370 - Methods and Definitions for Mechanical Testing of Steel Products.
B209 - Specification for Aluminum Alloy Sheet and Plate.
B211 - Specification for Aluminum Alloy Rolled and Cold Finished Rod and Bar.
B285 - Aluminum and Aluminum Alloy Welding Rods and Bare Electrodes.
B316 - Specification for Aluminum Alloy Rivet and Cold Heading Wire and Rods.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

American Association of State Highway Officials (AASHO) Publications:

M190 - Specifications for Bituminous Coated Corrugated Metal Culvert Pipe and Pipe Arches.
M196 - Specifications for Corrugated Aluminum Alloy Culvert Pipe.
M197 - Specifications for Corrugated Aluminum Alloy Pipe Underdrains.
M211 - Specifications for Helically Corrugated Aluminum Alloy Culvert Pipe.
M219 - Specifications for Aluminum Alloy Structural Plate for Pipe, Pipe Arches, and Arches.

(Application for copies should be addressed to the American Association of State Highway Officials, 341 National Press Building, Washington, D.C., 20004.)

American National Standards Institute (ANSI):

ANSI B18.2.2 - Square and Hex Nuts.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York, 10018.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Classes I, II, and III pipe.

3.1.1 Materials specifications. Pipe, pipe arches, and arches shall be fabricated from aluminum alloy sheets and plates conforming to the applicable specifications designated in Table I. At least one sample shall be taken by the producer from each group of ingots of the same alloy poured simultaneously from the same source of molten metal and analyzed to determine conformance to applicable specifications in Table I. Ingots not conforming to that specification shall be rejected. Complete ingot analysis records shall be available to the purchaser at the producer's plant.

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TABLE I.

Items	Use	Alloy	Specifications ASTM	Specifications AASHO
Sheets - Classes I and II		Alclad 3004-H34	B209	M196
Sheet and Plate - Class III, structural plate		5052-H141	B209	M219
Rivets - Classes I and II		6053-T4	B316	M196
Bolts (aluminum) - Class III		6061-T6	B211	
Bolts (steel) - Class III and all Coupling Bands			(A307 (A325 or (A449	
Coupling band angles or lugs extruded		6063-T6	B221	M196
Welding electrodes		ER-4043 or ER-5356	B285	

3.1.2 Corrugation configurations. Corrugations shall form smooth curves and tangents. The radii of curvature of the corrugations shall be at least one-half the depth of the corrugation. The crests and valleys of annular corrugations shall form circumferential rings (complete or partial) about the axis of the pipe. Helical corrugations shall form helices about the pipe axis and the direction of crests and valleys shall be not less than 45° from the longitudinal axis of the pipe. The series identification and dimensions of corrugations for the three classes of pipe shall be in accordance with Table II. The depth of corrugations shall not underrun those specified in Table II by more than 5 percent.

Table II. Series Designations, Corrugation Dimensions, in inches

Class	Series	Depth	Pitch Minimum	Pitch Maximum	Diameter Range Inches	Specifications
I - Annular	A	1/2	2-1/4	2-3/4	8 to 84	AASHO M196
	B	1	5-3/4	6-1/4	30 to 96	
II - Helical	A	1/2	1-7/8	2-3/4	12 to 84	M211
	C	1/4	1-3/8	1-7/8	6 to 10	M197
III - Annular	D	2-1/2	8-5/8	9-3/8	78 to 180	M219

3.1.3 Perforations for underdrains. When specified (see 6.2) perforations shall be arranged in two groups of longitudinal rows, one group placed symmetrically on each side of a non-perforated bottom segment corresponding to the flow line of the pipe. Within each group the rows of perforations shall be spaced transversely 1 or 1-1/2 inches center to center and in each row the perforations shall be located either on the inside crests, or on one of the neutral sides of all corrugations except that perforations are not required within 6 inches of each end of each length of pipe or in the corrugations where circumferential seams are located. The perforations shall have a diameter of not less than 1/4 inch and not more than 3/8 inch. When bituminous coating is specified for perforated pipe the clear diameter of each perforation after coating shall be not less than 75 percent of the diameter of perforation before coating. The minimum number of longitudinal rows of perforations and the minimum width of unperforated bottom segment shall be as shown in Table III. Perforations may be specified only for the classes and diameters of pipe shown in Table III.

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TABLE III. Perforation Details

Nominal Internal Diameter Inches	Class of Pipe	Number of longitudinal rows of perforations in pipe (2 groups) - Minimum	Width of unperforated bottom segment, minimum - inches
6	II	4	4-1/2
8	I and II	4	7
10	I and II	4	9
12	I and II	6	9-1/2
15	I and II	6	13
18	I and II	6	16-1/2
21	I and II	6	20

3.1.4 Pipe Designations. The diameter, number, and length of sections, and total length required shall be specified (see 6.2). Perforations shall also be specified when required.

3.1.5 Thickness of Aluminum Alloy Sheet and Plate. Thickness shall be as specified (see 6.2). Thickness and tolerance thereon shall be in accordance with Table IV.

TABLE IV. Thickness and Tolerance

	Total thickness, nominal, inches	Nominal Gage 2/	Tolerance under nominal 1/
Plate	0.25	-	0.020
	0.225	-	0.018
	0.20	-	0.016
	0.175	-	0.016
	0.150	-	0.014
	0.125	-	0.012
	0.100	-	0.007
	0.090	-	0.007
Sheet	0.164	8	0.009
	0.135	10	0.009
	0.105	12	0.008
	0.075	14	0.007
	0.060	16	0.007
	0.048	18	0.006

1/ No limit on overthickness

2/ For informational purposes only

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3.2 Class I, Series A and Series B Pipe.

3.2.1 Fabrication. Pipe furnished under this class shall be of riveted construction. Resistance spot welded construction may be used in lieu of rivets for Series A pipe of 0.06 and 0.075 inch thickness in diameters up to 36 inches when specified (in 6.2).

3.2.1.1 Riveted construction. Aluminum alloy rivets conforming to ASTM B316, Alloy 6053-T4 shall be used. All rivets shall be driven cold in such a manner that the sheets will be drawn tightly together throughout the entire lap. They shall have a neat, workmanlike, and full upset head of a form acceptable to the contracting officer, shall be driven without bending, and shall completely fill the holes. The centers of 5/16 inch and 3/8 inch diameter rivets shall not be closer than twice their diameter to the edge of sheets and centers of 1/2 inch diameter rivets shall not be closer than 1-1/2 times their diameter to the edge of sheets.

3.2.1.2 Resistance spot welded construction. Spot welds shall be made in such a manner that the sheets will be drawn tightly together throughout the entire lap. The weld diameters shall be at least three times the sheet thickness on all welds and the center of a spot weld shall be not less than its radius plus 1/4 inch from the edge of the sheet. Spot welds in longitudinal seams shall be located in the valley, crest, or tangent of each corrugation and in circumferential seams the spacing shall be the same as specified for rivets in 3.2.2.2.

3.2.1.2.1 Quality and procedure approval. Before welding production is permitted each welding machine and operator shall be qualified by destructive shear tests on coupons, supplemented by a peel test, as specified in 4.4.3.1 and 4.4.3.2. Aluminum clad sheets do not normally require preparation for welding but when they have become water stained they shall be mechanically and chemically prepared prior to spot welding. Discoloration of spot weld surfaces shall not be cause for rejection.

3.2.1.2.2 Shear strength. Single shear strength of spot welds shall conform to Table V.

TABLE V. Spot weld shear strength 1/

Corrugated sheet thickness (inches)	Single shear strength, minimum (pounds)
0.060	1,200
0.075	1,500
0.105	-
0.135	-
0.164	-

1/ Resistance spot weld shear strengths will result in lower longitudinal seam strength values and lower fill heights than in riveted or helical fabrication, as reflected in Fill Height Table IXa.

3.2.2 Series A pipe. Rivet sizes shall be not less than 5/16 inch diameter for 0.060 inch (16 gage) and 0.075 inch (14 gage) thickness of material and not less than 3/8 inch diameter for 0.105 inch (12 gage) material and thicker.

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3.2.2.1 Longitudinal seams. Longitudinal seams of pipe less than 42 inches in diameter shall be riveted with at least one rivet in the valley of each corrugation. The longitudinal seams of pipe 42 inches and larger in diameter shall have two rivets in the valley of each corrugation (double riveted) and the nominal distance between center lines of the two rivet rows shall be at least 3 times the rivet diameter.

3.2.2.2 Circumferential seams. Shop riveted circumferential seams shall have a rivet spacing not greater than 6 inches except that 6 rivets will be sufficient for 12-inch diameter pipe.

3.2.2.3 Perforations. Perforations when specified (see 6.2) shall conform to 3.1.3 and Table III.

3.2.3 Series B pipe. Rivets conforming to and driven as specified in 3.2.1.1 shall be used. Size of rivets shall be not less than 3/8 inch diameter for pipe diameters 30 inches to 42 inches, inclusive, and not less than 1/2 inch diameter for pipe diameters 48 inches and larger.

3.2.3.1 Longitudinal seams. All longitudinal seams in series B pipe shall be double riveted with 2 rivets in the valley and in the crest of each corrugation. The distance between center lines of the two rivet rows shall be not less than 1-1/2 inches.

3.2.3.2 Circumferential seams. All circumferential seams shall have a rivet spacing of not greater than 6 inches for rivets of sizes specified in 3.2.3.

3.2.4 Coupling bands, Class I pipe. Field joints for pipe sections shall be made with coupling bands which will provide the circumferential and longitudinal strength required to preserve the pipe alignment and to prevent separation of pipe sections, thereby minimizing infiltration of fill material. Coupling bands shall be made of the same alloy material as the pipe, shall have corrugations or dimple projections that mesh with the corrugations of the pipe sections to be connected, and shall be formed to fit the shape specified for those sections. The bands shall be not more than one tubular thickness (see Table IV) lighter than that of the pipe sections to be connected, but never less than 0.048 inch material. The bands shall be fabricated to lap on an equal portion of each of the sections connected.

3.2.4.1 Coupling bands with dimple projections. Bands with dimple projections shall have projections conforming substantially to the shape and depth of the pipe corrugations and shall be in circumferential rows with not less than seven projections per row. The bands may be used on Series A, Class I or Class II, or a combination thereof, and shall conform to Table VI. Bolts and nuts shall conform to 3.2.4.2.

TABLE VI. Requirements, pipe coupling bands with projections

Diameter	Width bands	Total rows of projections, and series		Size of connecting angles	Number of 1/2 inch diameter bolts
Inches	Min. inches	Min.	Series	Inches	Min.
12 to 54	10-1/2	2	A	2 by 2 by 3/16	2
60 to 84	16-1/4	4	A	2 by 2 by 3/16	3

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3.2.4.2 Series A pipe bands. Bands with corrugations shall be not less than 7 inches wide for pipe 8 inches to 30 inches in diameter, not less than 12 inches for pipe diameters 36 inches to 60 inches, inclusive, and not less than 24 inches for pipe diameters 66 inches and larger. The bands shall have 2 inch by 2 inch by 3/16 inch aluminum alloy connection angles conforming to ASTM B221, alloy 6063-T6, riveted securely to the band ends with rivets conforming to 3.2.2 for size and material. The bands shall be installed as specified in 3.2.4 and tightened in place by 1/2 inch galvanized steel bolts through the vertical legs of connection angles. The bolts shall conform to ASTM A307 galvanized in accordance with ASTM A153. Bands 7 inches wide shall have 2 bolts; 12-inch bands, 3 bolts; and 24-inch bands, 4 bolts. Bands with projections shall conform to 3.2.4 and Table VI.

3.2.4.3 Series B pipe bands. Coupling bands for pipe diameters of 30 inches to 60 inches, inclusive, shall be 18 inches wide and for diameters 66 inches and greater shall be 24 inches wide. Bands shall be connected at the ends by lugs formed and punched from 2 inch by 2-1/2 inch by 1/4 inch extruded angles conforming to ASTM designation B221, Alloy 6063-T6. The lugs shall be securely riveted to the bands with 3/8-inch diameter aluminum rivets, or by welding. Welding shall consist of a 3/16-inch fillet weld along each end and along the toe of each lug. Bands shall be installed in conformance with 3.2.4 and tightened in place by 1/2-inch diameter bolts through the vertical legs of connection angles. The bolts shall conform to ASTM A307, galvanized in accordance with ASTM A153. Bands 18 inches wide shall be connected by 3 bolts and bands 24 inches wide by 4 bolts.

3.2.4.4 Other coupling bands. Other equally effective coupling bands or methods for connecting the pipe sections may be used if approved by the contracting officer (see 6.2).

3.2.5 Shapes - Class I pipe.

3.2.5.1 Shape 1. Pipe furnished as Shape 1 shall be of full circle shape. Shape 1 may be specified with Series A and Series B corrugations.

3.2.5.2 Shape 2. Pipe furnished as Shape 2 shall be fabricated by reforming circular pipe to an approximate ellipse with the vertical diameter approximately 5 percent greater than the diameter of a corresponding circular shape. This shape also may be specified for Series A and Series B corrugations.

3.2.5.3 Shape 3. This shape shall be fabricated by reforming a circular pipe to a multi-centered one having an arch shaped top with a slightly convex curved integral bottom. Alternately this shape may be fabricated by curving individual sheets to form circular arcs as required for this shape. The lapped longitudinal seams shall be staggered so as to alternate on each side of the center of arch top by approximately 15 percent of the periphery. Pipe of this shape shall conform to the dimensional requirements in Table VII. A tolerance of plus, or minus, 1 inch is permissible for dimensions of span, rise, corner radius, and dimension B. This shape is fabricated only with Series A corrugations and when specified (see 6.2) the required span and rise must also be specified.

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TABLE VII.

Dimensional Requirements, Shape 3,
Class I and Class II, Series A, Pipe

Span	Rise	Diameter Equivalent Round	Corner Radius	Dimension B
		in inches		
18	11	15	4	4-1/2
22	13	18	4	4-3/4
25	16	21	4	5-1/4
29	18	24	4-1/2	5-1/2
36	22	30	5	6-1/4
43	27	36	5-1/2	7
50	31	42	6	8
58	36	48	7	9-1/4
65	40	54	8	10-1/2
72	44	60	9	11-3/4

3.2.6 Pipe diameters. The average inside diameter of circular pipe shall not vary more than one percent or 1/2 inch, whichever is greater, from the specified nominal diameter, when measured between inside crests of pipe.

3.3 Class II pipe.

3.3.1 Pipe fabrication. Pipe furnished under this class shall be Series A or Series C with helical corrugations conforming to 3.1.2 and Table II. Series C shall be fabricated into Shape 1 only and Series A in Shape 1, 2, or 3. Performations when specified (see 6.2) shall conform to 3.1.3 and Table III.

3.3.2 Seams. Class II pipe shall have a lock seam or a continuous welded seam paralleling the corrugations and extending continuously from end to end of each length of pipe. Either type of seam shall be so fabricated that the nominal diameter or shape of the pipe will not be affected nor will a plane of weakness be created that will appreciably lessen the load-carrying capacity of the pipe. Where welded seams are furnished, the manufacturer shall certify that the welds have been tested and found satisfactory. Such tests shall be made at the beginning of each shift and also when a change in metal thickness is made. The tests shall be acceptable to the purchaser.

3.3.3 Coupling bands. Field joints in Class II pipe shall be made with the same aluminum alloy as the pipe and shall have corrugations or dimple projections that will mesh with the corrugations of the pipe sections to be connected. The coupling bands shall be formed to fit the pipe shape specified and shall not be less than 7 inches wide for pipe diameters 6 inches to 30 inches, inclusive, not less than 12 inches wide for pipe diameters 36 inches to 60 inches, inclusive, and not less than 24 inches wide for pipe diameters 66 inches and larger. The thickness of such bands shall not be more than one tabular thickness (Table IV) lighter than that of the pipe sections to be connected but never less than 0.048 inch. The bands shall be one-piece or two-piece as specified (see 6.2) and shall be so fabricated as to lap on an equal portion of each of the pipe sections to be connected. The bands shall have aluminum alloy connection

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angles or aluminum lugs securely riveted to the ends thereof by aluminum rivets and connected by 1/2-inch galvanized ASTM A307 steel bolts all in accordance with 3.2.4.2 Bands with dimple projections shall conform to 3.2.4.1, Table VI.

3.3.3.1 Coupling bands with slip seams. For pipe diameters 12 inches to 36 inches, inclusive, one-piece coupling bands utilizing a slip seam may be used. This device provides for coupling band expansion by mechanically slipping the lock seam so that the coupler may be slipped over the ends of the pipe sections to be connected. The coupler is drawn snug over the pipe by reversing the method of slipping the seam and is then tightened to prevent further slippage. Mechanical slipping of seams is effected by means of a 1/2 inch bolt with double reverse threads used in conjunction with extruded aluminum lugs tapped to receive the 1/2 inch threaded bolt and welded to the band, one on each side of the helical seam.

3.3.3.2 Other coupling bands. Other equally effective coupling bands or methods for connecting the pipe sections may be used when specified in 6.2 or if approved by the contracting officer.

3.3.4 Shapes - Class II pipe.

3.3.4.1 Shape 1. Pipe furnished as Shape 1 shall be of full circle shape and may be specified with Series A or Series C corrugations.

3.3.4.2 Shape 2. Pipe furnished in this shape shall be fabricated by reforming circular pipe to an approximate ellipse with the vertical diameter approximately 5 percent greater than that of the circular shape. This shape may be specified only with Series A corrugations.

3.3.4.3 Shape 3. This shape shall be fabricated by reforming a circular pipe to a multi-centered one having an arch shaped top with a slightly convex curved bottom and may be specified with Series A corrugations. Pipe of this shape shall conform to the dimensional requirements in Table VII and span and rise shall be specified. A tolerance of plus or minus one inch is permissible for dimensions of span, rise, corner radius, and dimension B.

3.3.5 Pipe diameters. The average inside diameter of circular pipe shall not vary more than 1 percent or 1/2 inch, whichever is greater, from the specified nominal diameter when measured between inside crests of pipe.

3.4 Class III pipe.

3.4.1 Fabrication. Class III shall be structural plate pipe and structures field assembled with bolted lap joint construction.

3.4.2 Bolts. Bolts and nuts for connecting plates shall be 3/4 inch diameter and shall conform to the dimensions of USA Std. B18.2.2 except that the bearing faces of bolt heads and nuts shall be symmetrically shaped to a 22 degree chamfer with the maximum bolt head and nut thickness not less than the minimum specified for regular hexagon bolts and nuts. Regular hexagonal bolts and nuts may be used in lieu of the above shaped heads and nuts if washers shaped to fit the corrugations are furnished. There shall be at least 5-1/3 bolts per foot of longitudinal seam and the bolt spacing on circumferential seams shall not exceed 10 inches. The number of bolts of each size and length furnished shall be 5 percent plus 2 in excess of the theoretical number required for field erection of the structure or structures. Bolt lengths shall be such as to result in at least 1-1/2 threads past the nut when tightened in place.

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3.4.2.1 Aluminum alloy bolt and nut material. Aluminum alloy bolt and nut material shall conform to the requirements of ASTM B211, Alloy 6061-T6. Nuts shall be factory prelubricated with a suitable wax compound.

3.4.2.2 Steel bolt and nut material. Steel bolt and nut material shall conform to the requirements of ASTM A449 and ASTM A563, grade C, respectively, and shall be hot-dip galvanized in accordance with ASTM A153. ASTM A325 bolts may be used in lieu of A449 bolts.

3.4.3 Structural plates. Plates furnished under Class III shall consist of structural units of corrugated aluminum plates formed to provide lap seams. Single plates shall weigh not more than 250 pounds. The plates at longitudinal and circumferential seams shall be fabricated for connection by bolts as specified in 3.4.2. Joints shall be staggered so that not more than three plates come together at any one point. Each plate shall be curved to the proper radius to form one or more circular arcs as required to form the shape specified so that cross-sectional dimensions of the finished structure will be as indicated on the procurement plans.

3.4.3.1 Bolt holes. Bolt holes shall be punched so that all plates having like dimensions, curvature, and same number of bolts per foot of seams shall be interchangeable. Unless otherwise specified the bolt holes along the plate edges that will form longitudinal seams in the finished structure shall be in two rows 1-3/4 inches apart and located in the valley and in the crest of each corrugation. The minimum distance from the center of bolt holes to edge of plates shall be not less than 1-3/4 times the bolt diameter. The diameter of bolt holes in longitudinal seams, excepting those at plate corners, shall not exceed the bolt diameter by more than 1/8 inch. The diameter of holes in circumferential seams, including those at plate corners, shall not exceed the bolt diameter by more than 1/2 inch and the average of the diameters on the major and minor axes shall not exceed the bolt diameter by more than 1/4 inch.

3.4.3.2 Plate edges and identification. Plates for forming skewed or beveled ends of structures shall be accurately cut to form the specified final shape. Plates shall be saw cut, not burned, and shall present a workmanlike finish free from notches or gouges. Legible identification symbols shall be placed on each part plate to designate its proper position in the finished structure as indicated on the approved erection drawings.

3.4.4 Shapes. Class III structures shall be furnished in shapes 2, 3, and 4 as specified (see 6.2).

3.4.4.1 Shape 2. Structures furnished in this shape are designated as structural plate pipe. The plates shall be curved so that when erected they will form an ellipse with the long axis vertical. The long axis shall be not more than 5 percent, or 10 percent (whichever is specified, see 6.2), greater than the nominal diameter of a corresponding circular pipe.

3.4.4.2 Shape 3, pipe arch. Structural plates for this shape shall be curved to form a cross section made up of four circular arcs tangent to each other at their junctions and symmetrical about the vertical axis. The top arc shall be joined at each end to the bottom arc by a corner arc having an inside radius of 28.8 inches and a 92 degree subtended angle. The radius and angle remain constant for all sizes of pipe arches. Dimensions shall be shown on procurement plans and span and rise shall be specified (see 6.2).

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3.4.4.3 Shape 4, arch. Structural plates for arches shall be curved to form arches of circular arc cross section when erected. All dimensions shall conform to the procurement plans and span and rise shall be specified (see 6.2). Increased headroom or waterway area may be secured by adding to each side of the arch a plate tangent to the lower end thereof and bolted thereto. Tangent sections shall be specified (see 6.2) when required. All details shall be shown on the procurement plans.

3.4.4.4 Bearing details. Bearing details shall be as shown on the procurement plans.

3.5 Coatings - bituminous. Coatings A to E shall be an asphalt conforming to AASHO M190 (see 4.4.4.1).

3.5.1 Coating A, fully bituminous coated. The pipe shall be uniformly coated with bituminous material inside and outside to a minimum thickness of 0.05 inch measured on the crest of the corrugations (see 4.4.4.2).

3.5.2 Coating B, half bituminous coated, paved invert. The pipe shall be coated for approximately the lower half of the circumference (bottom of pipe, installed) inside and outside as specified for Coating A, and in addition the bituminous material shall be so applied as to form a smooth pavement in the invert (inside bottom of pipe installed) filling the corrugations for at least 25 percent of the circumference for shapes 1 and 2, and 40 percent for shape 3. The invert pavement shall have a minimum thickness of one-eighth (1/8) inch above corrugation crests except where the upper edges thereof intersect the corrugations (see 4.4.4.2).

3.5.3 Coating C, fully bituminous coated, paved invert. The pipe shall be fully bituminous coated as specified for coating A and in addition the invert shall be paved as specified for coating B.

3.5.4 Coating D, paved invert only. The pipe shall have a paved invert only, as specified for coating B.

3.5.5 Coating E, fully bituminous coated, fully paved. The pipe shall be bituminous coated as specified for coating A and in addition the bituminous material shall be so applied as to form a smooth pavement over the complete inner surface of pipe. This pavement shall have a minimum thickness of 1/8 inch over the corrugation crests (see 4.4.4.2).

3.6 Workmanship. Pipe shall show careful workmanship in all particulars and shall be free from the following defects:

(a) Undue deviation from true shape.
(b) Uneven laps.
(c) Variations from a reasonably true centerline.
(d) Ragged or diagonally sheared edges.
(e) Loose, unevenly lined, or spaced rivets or bolts.
(f) Poorly formed rivet heads.
(g) Illegible brand.
(h) Dents or bends other than corrugations.
(i) Cracks, skips, or deficient welds in machine welding.
(j) Poorly formed lock seams and damaged lock seam metal.
(k) Inadequate, improperly applied, cracked, or loose asphalt coating.

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4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all pipe and structural sections of the same class, series, shape, and coating offered for delivery at the same time.

4.2.2 Sampling for dimensional and visual examination of end item. Sampling of corrugated pipe for these characteristics shall be in accordance with inspection level S-4.

4.2.3 Sampling for tests. Samples for tests shall be selected in accordance with inspection level S-2.

4.3 Examination. The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 End item. Pipe shall be examined in accordance with Table VIII. The AQL's for defects listed in Table VIII shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE VIII. Classification of defects, end items.

Defects	Major	Minor
Class, series, shape, or coating not as specified	x	
Dimensions, including diameter, length, and thickness of metal not as specified	x	
Uneven laps		x
Deviation from a true centerline		x
Ragged or diagonally sheared edges		x
Loose, unevenly lined, or spaced rivets or bolts	x	
Poorly formed rivet heads	x	
Illegible brand marking		x
Dents or bends other than corrugations		x
Cracks, skips, or deficient welds in machine welding	x	
Lock seams poorly formed or damaged	x	
Asphalt coating or paving significantly cracked or loose	x	

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4.3.1.1 Pitch of corrugations. Measure pitch at right angles to direction of corrugations.

4.3.1.2 Thickness of sheet and plate. Measure thickness normal to the tangent of corrugations.

4.3.1.3 Measurements, classes I and II pipe. Measure all dimensions of Classes I and II pipe arches from the inside crests of corrugations.

4.3.1.4 Measurements, dimension "B". Measure dimension "B" in Table VII of Classes I and II pipe arches vertically from the lowest part of the base to a horizontal line drawn across the widest part of the arch.

4.4 Test methods.

4.4.1 Chemical property limits and mechanical property limits. Component parts used for corrugated aluminum pipe shall be tested in accordance with the applicable ASTM or AASHO Specifications listed in 3.1.1, Table I.

4.4.2 Fasteners.

4.4.2.1 Tests, aluminum bolts. Conduct tensile, bending, and hardness tests in accordance with ASTM B211, alloy 6061-T6 for aluminum bolts.

4.4.2.2 Testing, steel bolts. Conduct tensile, bending, and hardness tests in accordance with ASTM A370 specification for steel bolts.

4.4.3 Resistance spot welds.

4.4.3.1 Specimens. For qualifying welds and welding procedure approval prepare three specimens for single spot shear tests and one specimen for a peel test. Each coupon (specimen) for the spot weld shear test shall be prepared by lapping two strips of corrugated sheet, representative of the materials to be welded, each strip 1-1/2 corrugations wide by 6 inches long and joining them together by a single spot weld. The size of the spot weld and width of lap shall be the same as those used in production. The average of the three tests shall be considered the shear value which shall conform to Table V. If the average value is lower than that specified, the welding machine shall be readjusted and shear tests repeated until the shear values so obtained conform. For the peel test a specimen having two strips of corrugated sheets, each six corrugations wide, and joined with at least five spot welds shall be prepared. After welding, the strips shall be forced apart (peeled) and the weld size measured. The diameter of weld nugget or fused zone shall be at least three times the thickness of one sheet (strip) for all welds. The peel test shall be made at least once a day, and at changes of machine adjustment, or change in thickness of sheet.

4.4.3.2 Procedure. Mount each specimen in turn on a standard calibrated tensile testing machine and test to destruction in tension (see 3.2.1.2.1 and Table V).

4.4.3.3 Frequency of tests. Perform tests at each change in machine adjustment, operator, thickness of sheet, or brand of sheet material.

4.4.4.1 Coatings A to E. Conduct tests in accordance with AASHO M190 (see 3.5).

4.4.4.2 Measure thickness of coating at an exposed end or with a magnetic or scratch gage (see 3.5.1 to 3.5.5).

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4.5 Rejection.

4.5.1 Tests. A lot shall be rejected for failure to meet the test requirements of this Specification when tested in accordance with 4.4.

4.5.2 Conduct retests using at least two specimens.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking of the pipe shall be inspected to determine conformance with the applicable requirements of Section 5 of this Specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Bolts. Bolts or other fastening devices shall be packaged in fiberboard boxes conforming to PPP-B-636 (class weather-resistant).

5.1.2 Level C.

5.1.2.1 Bolts. Bolts or other fastening devices shall be packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Bolts and coupling bands. Bolts and other fastening devices packaged in accordance with paragraph 5.1 and coupling bands, shall be packed in cleated plywood boxes conforming to PPP-B-601, overseas type.

5.2.1.2 Class I and class II pipe. Pipe up to 24 inches in diameter shall be packed in crates conforming to PPP-C-650, type III, style B. Flat steel strapping shall conform to QQ-S-781, type I, class B. Pipe over 24 inches in diameter shall be shipped loose.

5.2.1.3 Class III pipe. Pipe shall be packed in accordance with the level A packing requirements of MIL-P-236 for type I culvert pipe.

5.2.2 Level B.

5.2.2.1 Pipe, bolts, and coupling bands. Pipe, bolts, and coupling bands shall be packed as specified for Level A except that boxes shall be domestic type and strapping need not be class B.

5.2.3 Level C.

5.2.3.1 Pipe, bolts, and coupling bands. Pipe, bolts, and coupling bands shall be packaged in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.3 Markings.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Pipe and structures covered by this specification are intended for use as culverts, storm drains, underdrains, conduit, and miscellaneous applications. Where additional protection is required because of unusually corrosive or erosive conditions, a coating type should be specified for both pipe and coupling bands.

6.2 Ordering data. Purchasers shall select the preferred options permitted herein and include them with the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class, series and shape (see 1.2.1).
(c) Coating (see 1.2.2 and 6.3.1).
(d) Class I, Series A, riveted or spot welded fabrication, if one is preferred (see 3.2.1).
(e) Classes I and II, perforations and pattern, if desired, for drainage (see 3.1.3 and Table III).
(f) Classes I, III, and III, metal thickness (see 3.1.5 and Table IV).
(g) Classes I, II, and III, number, length, and diameter of sections, and total lengths required (see 3.1.4 and 6.4).
(h) Classes I, II, and III, Shape 3, specify the span and rise required (see 3.2.5.3, 3.3.4.3, and 3.4.4.2).
(i) Class III, Shape 2, specify 5 percent or 10 percent elongation if one is preferred (see 3.4.4.1).
(j) Classes I and II, specify type, class, and number of coupling bands, if one is preferred (see 3.2.4 and 3.3.3).
(k) Class III, Shape 4, specify dimensions, span, rise, and bearing details for arches (see 3.4.4.3).
(l) Size and lot if different from 4.2.1.
(m) Selection of applicable level of packing required (see 5.1).

6.3 Coatings

6.3.1 Coating availability. Coatings A to E are available for use with Class I and Class II pipe. Coating A only may be specified for perforated pipe.

6.4 Length. The length of pipe and structural sections shall be subject to modification when necessary to conform to transportation regulations.

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6.5 Design considerations. Fill Height Tables IX, IXa, X, Xi, XII, and XIII give maximum fill heights for the various classes, series, shapes, and thicknesses of metal and are intended only as a guide. For design and installation information consult the following publication dated June 1966: 1/

Corrugated Metal Pipe Culverts
Structural Design Criteria and
Recommended Installation Practices.
U. S. Department of Commerce
Bureau of Public Roads 2/

1/ This publication is being revised and will be available early 1970.

2/ Now in the U. S. Department of Transportation (DOT).

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TABLE IX. Fill Heights 1/
Classes I and II, Series A, Corrugated Aluminum Pipe
2-2/3-inch by 1/2-inch Corrugations
Riveted or Helical Fabrication

Pipe Diameter (inches)	Minimum Cover Top of Pipe to Top of Subgrade (inches)	Maximum Fill Heights Above Pipe Top (in feet) For Sheet Metal Thickness (in inches)									
		0.06		0.075		0.105		0.135		0.164	
		Fabrication Shapes 2/									
		1	2	1	2	1	2	1	2	1	2
12	12	45	-	45	-	77	-	-	-	-	-
18	12	28	-	30	-	36	-	43	-	49	-
24	12	22	-	23	-	25	-	28	-	31	-
30	15	18	18	18	18	21	31	23	33	24	34
36	15	15	15	15	15	19	26	20	27	21	28
42	15	-	-	18	25	19	36	19	38	20	40
48	15	-	-	-	-	18	24	18	31	19	38
54	24	-	-	-	-	17	17	18	21	18	27
60	24	-	-	-	-	-	-	16	16	18	20
66	24	-	-	-	-	-	-	12	12	15	15
72	36	-	-	-	-	-	-	-	-	11	11

TABLE IXa. Fill Heights 1/
Class I, Series A, Corrugated Aluminum Pipe
2-2/3-inch by 1/2-inch Corrugations
Spot Welded Fabrication

Pipe Diameter (inches)	Minimum Cover Top of Pipe to Top of Subgrade (inches)	Maximum Fill Heights Above Pipe Top (in feet) For Sheet Metal Thickness (in inches)									
		0.06		0.075		0.105		0.135		0.164	
		Fabrication Shapes 2/									
		1	2	1	2	1	2	1	2	1	2
12	12	26	-	33	-						
18	12	18	-	22	-						
24	12	14	-	16	-	not fabricated					
30	15	11	11	13	13						
36	15	9	9	11	11						

1/ For H20 loading and fill material at 120 lbs./cu.ft. Minimum recommended compaction; 85 percent Standard Proctor Density.

2/ Shape 1 - Circular.
Shape 2 - Elongated; fabricated in diameters 30 inches and larger only.

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TABLE X. Fill Heights 1/
Classes I and II, Series A, Corrugated Aluminum Pipe Arches
2-2/3-inch x 1/2-inch Corrugations
Riveted, Spot Welded 2/, or Helical Fabrication

Dimensions Span (inches)	Rise	Corner Radius (inches)	Minimum Cover Top of Pipe to Top of Subgrade (inches)	Minimum Thickness of Metal (inches)	Maximum Fill Heights in feet Above Top of Pipe for Corner Bearing Pressure in tons/sq.ft.	
					2 tons	3 tons
18	11	4-3/4	18	0.06	15+	-
22	13	4-3/4	18	0.06	14	-
25	16	4-1/2	18	0.06	12	15+
29	18	4-1/2	18	0.06	10	15
36	22	5	18	0.06	9	14
43	27	5-1/2	18	0.075	9	13
50	31	6	18	0.105	8-	12
58	36	7	18	0.135	8-	12
65	40	8	18	0.135	8	12
72	44	9	18	0.164	8	12

1/ For H20 loading and fill material at 120 lbs./cu ft. Minimum recommended compaction; 85 percent Standard Proctor Density.

2/ Fill heights for spot welded pipe arches of spans up to 36 inches only and for corner bearing pressure of 2 tons/sq. ft.

TABLE XI. Fill Heights 1/
Class I, Series B, Corrugated Aluminum Pipe
6-inch by 1-inch Corrugations
Riveted Fabrication

Pipe Diameter (inches)	Minimum Cover Top of Pipe to Top of Subgrade (inches)	Maximum Fill Heights Above Pipe Top (in feet)									
		For Sheet Metal Thickness (in inches)									
		0.06		0.075		0.105		0.135		0.164	
		Fabrication Shapes 2/									
		1	2	1	2	1	2	1	2	1	2
30	15	27	29	30	37	35	56	41	58	46	60
36	15	23	24	24	31	28	47	30	48	34	50
42	15	21	21	22	27	24	41	26	42	28	43
48	15	20	24	20	28	22	36	23	44	24	48
54	24	19	22	19	25	20	32	21	39	22	44
60	24	18	20	19	22	20	29	20	35	21	42
66	24	18	18	18	20	19	26	19	32	20	38
72	36	14	14	17	17	19	24	19	29	19	35
78	36	-	-	14	14	18	18	18	24	19	30
84	36	-	-	-	-	15	15	18	18	18	24
90	36	-	-	-	-	-	-	16	16	18	18
96	36	-	-	-	-	-	-	14	14	16	16

1/ For H20 loading and fill material at 120 lbs./cu.ft. Minimum recommended compation; 85 percent Standard Proctor Density.

2/ Shape 1 - Circular
Shape 2 - Elongated

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TABLE XII, Fill Heights 1/
Class III - Structural Aluminum Plate Pipe
9-inch by 2-1/2-inch Corrugations
Bolted Fabrication

Pipe Diameter (inches)	Minimum Cover Top of Pipe to Top of Subgrade (inches)	Maximum Fill Heights Above Pipe Top (in feet) For Sheet Metal Thicknesses of (in inches)							
		0.09	0.10	0.125	0.150	0.175	0.20	0.225	0.25
		Shape 2; 5 percent elongation							
78	12	17	20	27	34	41	46	51	55
84	12	16	19	25	32	38	42	47	51
90	12	15	17	22	29	35	40	44	48
96	12	14	16	22	27	33	37	41	44
102	12	13	15	20	26	31	35	39	42
108	24	12	14	19	24	29	33	37	40
114	24	-	14	19	23	27	31	35	38
120	24	-	13	17	22	26	30	33	36
126	24	-	12	16	21	25	28	32	34
132	24	-	12	16	20	24	27	30	33
138	24	-	-	15	19	23	26	29	32
144	24	-	-	14	18	22	25	27	30
150	24	-	-	14	18	21	24	26	29
156	24	-	-	12	16	21	23	25	28
162	24	-	-	-	16	19	22	24	26
168	24	-	-	-	15	19	21	23	26
174	24	-	-	-	-	18	21	23	25
180	24	-	-	-	-	17	20	22	24

1/ For H20 loading and fill material at 120 lbs./cu. ft.
Minimum recommended compation; 85 percent Standard Proctor Density.

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TABLE XIII, Fill Heights 1/
Class III, Structural Aluminum Plate Pipe Arches
9-inch by 2-1/2-inch Corrugations
Shape 3, Bolted Fabrication

Dimensions span	rise	Corner radius	Minimum Cover Top of Pipe to Top of Subgrade (in inches)	Minimum Thickness of Metal	Maximum Fill Heights (in feet) Above Top of Pipe for Corner Bearing Pressure in tons/sq.ft. 2 tons	3 tons
5' 11"	5' 4"	28.8	18	0.09	15+	-
6' 8"	5' 7"	28.8	18	0.09	15+	-
7' 4"	5' 11"	28.8	18	0.09	15	-
8' 0"	6' 2"	28.8	18	0.09	14	-
8' 7"	6' 6"	28.8	24	0.09	13	-
9' 0"	6' 8"	28.8	24	0.09	12	-
9' 4"	6' 10"	28.8	24	0.10	13	-
10' 0"	7' 1"	28.8	24	0.10	13	-
10' 5"	7' 3"	28.8	24	0.10	12	-
11' 2"	7' 6"	28.8	24	0.125	15	-
11' 8"	7' 10"	28.8	24	0.125	14	-
12' 2"	8' 0"	28.8	24	0.125	13	-
12' 10"	8' 3"	28.8	24	0.125	13	-
13' 7"	8' 7"	28.8	24	0.15	12	15+
14' 3"	8' 10"	28.8	24	0.15	11	15
14' 9"	9' 2"	28.8	24	0.175	10	15+
15' 3"	9' 4"	28.8	24	0.175	10	15+
16' 0"	9' 7"	28.8	36	0.20	10	15
16' 8"	9' 11"	28.8	36	0.225	10	14
16' 11"	10' 1"	28.8	36	0.25	9	14

1/ For H20 loading and fill material at 120 lbs./cu ft. Minimum recommended compation; 85 percent Standard Proctor Density.

MILITARY INTEREST:

Custodian

Army - ME
Navy - YD
Air Force - 82

CIVIL AGENCY INTEREST:

DOT-PR

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

WW-P-402C, PIPE, CORRUGATED (ALUMINUM ALLOY)

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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